FUNDAMENTALS OF COMPOSITE MATERIALS | 21MEO101T

WaveSense-CFRP

NOVELTY-FOCUSED TECHNICAL PROJECT DOSSIER

Temperature-Robust, Sensor-Efficient and Reliability-Aware Ultrasonic Guided-Wave Damage Assessment of Carbon Fiber-Reinforced Polymer Laminates

Proposed technical contribution: temperature-stability-aware sensor-path selection combined with missing-measurement robustness and calibrated, abstention-capable diagnosis.



SUBMISSION INFORMATION

Field	Details
Institution	SRM Institute of Science and Technology
Course / code	Fundamentals of Composite Materials / 21MEO101T
Department offering course	Mechanical Engineering
Student / registration no.	_____
Faculty guide / section	_____
Proposed project format	Software-only research and engineering prototype
Proposal version / date	Version 2.0 / 14 October 2026

STATUS: Proposed research and implementation plan. No model has yet been trained, and no experimental performance or novelty breakthrough is claimed.

Executive decision brief

This proposal upgrades the original WaveSense-CFRP concept from standard ultrasonic damage classification to a reliability-focused inspection framework. The project is fully digital: real signals were acquired by external laboratories, while our team will create original preprocessing, path-selection, analysis, model-evaluation and reporting software.

Proposed core contribution

- Temperature-aware path ranking: identify transmitter–receiver paths and excitation frequencies with strong reference-damage discriminability and comparatively low healthy-state thermal variability.
- Sensor efficiency and fault resilience: quantify classification performance using limited path budgets and when an entire transducer or selected measurements are unavailable.
- Reliability-aware decisions: calibrate model scores and issue Healthy / Reference damage / Inconclusive decisions rather than forcing unreliable classifications.
- Leakage-resistant experimental design: test held-out temperature intervals and, where feasible, unseen reference-damage positions with selection performed only on training data.

Faculty-ready abstract

Ultrasonic guided waves are widely investigated for condition monitoring of fiber-reinforced composite structures. Their propagation is sensitive not only to changes associated with structural discontinuities but also to environmental temperature, creating challenges for reliable damage detection. WaveSense-CFRP proposes a reproducible, software-only inspection framework using the publicly available Open Guided Waves (OGW) Dataset #2, which records multi-path ultrasonic responses of a carbon fiber-reinforced polymer (CFRP) laminate across controlled temperatures and reversible reference-damage conditions. The proposed method ranks measurement paths by joint damage sensitivity and healthy-state thermal stability, selects informative subsets under an explicit measurement budget, and evaluates supervised classifiers alongside signal-processing baselines. Robustness is assessed through held-out-temperature evaluation and simulated unavailable measurements, including transducer-level failures. Confidence calibration and an abstention mechanism identify inconclusive cases. The work will deliver an independently implemented algorithm, a transparent evaluation protocol, visual analytics and an interactive inspection dashboard. The aim is to determine whether temperature-aware sensor-path selection can improve the reliability-versus-measurement-efficiency trade-off; any improvement will be reported only after experimental validation.

Proposed outcomes - not promises

Research output	Evidence required
Temperature-aware ranking method	Explicit algorithm, reproducible code, train-only parameter selection
Comparison with existing baselines	Held-out metrics, ablations, uncertainty intervals
Path efficiency report	F1/recall/false-alarm rate against number of paths and unique sensors
Reliability module	Calibration plots, Brier score, risk–coverage trade-off
Digital engineering application	Live re-analysis of labeled public records and downloadable report

01 Academic context and course alignment

The supplied syllabus identifies the open-elective course 21MEO101T, “Fundamentals of Composite Materials,” with learning outcomes covering composite constituents, laminate mechanics, composite processing, testing methods and advanced composite applications. Unit 4 explicitly covers non-destructive inspection using ultrasonics, and Unit 2 addresses laminate behavior and damage mechanisms. [S1]

Unit	Relevance to WaveSense-CFRP
1 - Introduction	CFRP constituents, reinforcement architecture, matrix and interfaces.
2 - Mechanical analysis	Anisotropic laminated response; damage-related changes in mechanical wave propagation.
3 - Composite processing	CFRP plate fabrication, laminate integrity and relevance of manufacturing quality.
4 - Performance / quality	Direct connection to ultrasonic non-destructive testing and inspection methods.
5 - Advanced composites	Advanced fiber-reinforced materials and condition-monitoring applications.

02 Problem statement and research gap

A conventional classifier trained on ultrasonic signals under one set of conditions may exploit differences that are unrelated to damage, including temperature and recording-session artifacts. Using every path raises data processing demands and may over-represent redundant information. Missing transducer recordings can also degrade a system that assumes a complete sensor network. Furthermore, a forced binary prediction is not appropriate when confidence is low or the observation is outside conditions learned from training data.

Existing work and defensible contribution

The OGW Dataset #2 source paper already demonstrates temperature-sensitive signals and reports baseline-selection, PCA/ICA and adaptive-reference approaches. Related literature includes temperature-compensation research and uncertainty assessment for guided-wave diagnosis. Therefore, none of “ML detection,” “temperature compensation,” “sensor selection,” or “uncertainty” is individually claimed as a first-time invention. [R1, R3, R4]

CLAIM OF NOVELTY: A project-specific, independently implemented and ablated combination of temperature-stability-aware path ranking, sensor-failure stress testing and abstention-calibrated classification. Its incremental research value must be established by comparison, not assumed.

Research questions

1. RQ1. Does temperature-stability-aware path selection outperform random and conventional path selection at the same measurement budget on held-out temperature conditions?
2. RQ2. How does performance change when selected recording paths or all measurements involving one transducer become unavailable?
3. RQ3. Does confidence calibration with an inconclusive option reduce error among issued decisions, and at what coverage cost?
4. RQ4. How well does the method generalize to a reference-damage location that was not used for model selection or training?

Working hypotheses

H1: The proposed path ranking will improve the detection-versus-path-budget trade-off under selected held-out temperatures. H2: Robust training or path substitution will reduce degradation under simulated missing measurements. H3: An abstention threshold chosen solely on validation data will reduce error among accepted predictions while reducing coverage. These are testable hypotheses, not claimed results.

03 Technical background

Ultrasonic guided waves propagate within the boundaries of thin plate-like structures. In layered CFRP, the behavior is influenced by laminate architecture, anisotropy, frequency, temperature and geometry. A transmitter launches a tone burst; receivers observe complex waveforms containing multiple arrivals, scattered components and reflections. Damage proxies can alter amplitude, phase and time-of-flight, but temperature can change these quantities without any defect. [R1]

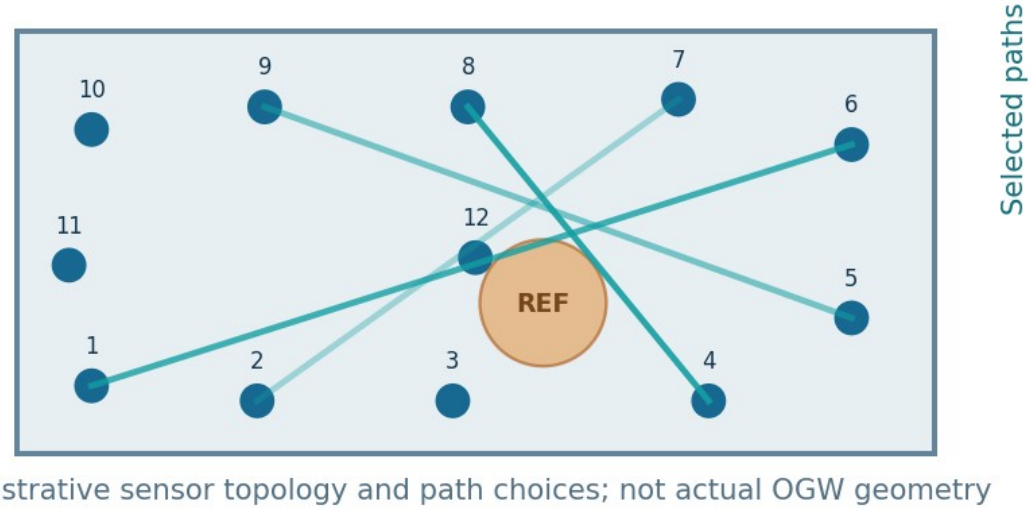


Figure 1. Conceptual sensor-path subset on a plate. This is a schematic created for the proposal; it does not reproduce the publisher's transducer coordinates.

Interpretation rules

- A detected signature is evidence of a changed guided-wave response under the modeled conditions, not proof of a specific internal failure mechanism.
- The OGW reference damage consists of a surface-mounted aluminum disc attached using tacky tape. It is reversible and must not be described as an experimentally confirmed internal delamination. [R1]
- A model validated on one specimen and controlled reference defects cannot be asserted to work on arbitrary real-world CFRP structures.

04 Dataset provenance and access

Primary dataset: Open Guided Waves (OGW) Dataset #2

Source: Moll, Kexel, Pötzsch, Rennoch and Herrmann, Scientific Data 6, 191 (2019), "Temperature affected guided wave propagation in a composite plate complementing the Open Guided Waves Platform." DOI: 10.1038/s41597-019-0208-1. Archive collection DOI: 10.6084/m9.figshare.c.4488089.v1. The official OGW site links to the publisher-hosted files. [R1, D1]

Published property	Description
Specimen	Quasi-isotropic HexPly CFRP laminate; 500 x 500 mm; 2 mm thick.
Laminate layup	[45/0/-45/90/-45/0/45/90]s, as reported by data authors.
Transducers	12 integrated piezoelectric discs; round-robin transmitter–receiver measurements.
Excitation	5-cycle Hann-windowed tone burst; 40–260 kHz at 20 kHz intervals.
Temperature	20–60 degrees C; 0.5-degree increments during controlled chamber cycles.
Healthy/reference states	Two intact temperature cycles; one cycle for each of four disc positions.
Damage positions	D04, D12, D16, D24; removable surface-mounted aluminum-disc proxy.
Raw data	HDF5 (.h5) inside timestamped folders; filenames identify carrier frequency.
Research use	Re-analysis and algorithm comparison with attribution to data creators.

CRITICAL DATA LIMITATION: Each damaged configuration belongs to a separate recorded cycle, and there is one principal specimen. Condition, temperature trajectory, day/cycle and location may be partially confounded. Treat record-level random splits as invalid evidence of real-world generalization.

Five published folder categories

- OGW_CFRP_Temperature_udam - intact baseline
- OGW_CFRP_Temperature_dam_D04 - proxy at D04
- OGW_CFRP_Temperature_dam_D12 - proxy at D12

- OGW_CFRP_Temperature_dam_D16 - proxy at D16
- OGW_CFRP_Temperature_dam_D24 - proxy at D24

Secondary sources and use constraints

Dataset	Appropriate use	Decision
OGW Dataset #1	Additional constant-temperature / artificial-reference-damage analysis.	Optional transfer check; do not pool naively.
NASA CFRP composites	Different fatigue and sensor experiment; possible later research extension.	Out of primary course scope.
Other experimental delamination benchmarks	Test whether methods transfer to real/internal-like damage.	Requires separate domain and label audit.

Dataset access checklist before modeling

5. Retrieve the official dataset catalogue and source article; store DOI, version, download date and file checksums.
6. Download a manageable subset and verify file readability, time axis, sampling rate, frequency tags and 12-channel relationships.
7. Count unique acquisitions, temperature-cycle identifiers, source files, transmitters, receivers and class labels.
8. Measure storage/memory requirements locally before deciding whether to use all frequencies and sensor paths.
9. Freeze the metadata manifest and split protocol before any supervised feature ranking.

05 Proposed data model and signal preprocessing

One canonical acquisition record

Each analysis unit will be linked to a complete measurement acquisition rather than treating each waveform from that acquisition as an independent statistical observation. Proposed metadata keys are listed below; exact native HDF5 fields will be reconciled after inspection.

Field	Purpose
source_file / source_hash	Reproducible provenance and integrity
condition / damage_position	Intact or D04/D12/D16/D24 reference-damage label
temperature_C / cycle_id / timestamp	Environmental condition and leakage control
frequency_kHz / transmitter / receiver	Frequency-specific measurement path ID
sampling_interval / waveform	Time-base interpretation and raw signal
acquisition_group	Split identifier linking correlated recordings

Preprocessing workflow

10. Ingest HDF5 data using h5py, validate metadata, and preserve read-only raw signals.
11. Remove malformed traces; verify finite sample values, channel lengths and monotonic time axes.
12. Remove or separately model the excitation/crosstalk region using a pre-declared physically justified time window.
13. Apply frequency-aware filtering and trace normalization using training-only parameters; examine effect of filtering by ablation.
14. Align signals and choose healthy references only from approved training/calibration pools, never by looking up an unknown test label.
15. Extract time-, frequency- and time-frequency-domain features; cache them with the preprocessing settings and software version.

Candidate features

Category	Feature examples / interpretation
Time domain	RMS, peak amplitude, energy, envelope, crest factor
Similarity	Normalized cross-correlation, residual energy, relative baseline change
Timing	Time-of-flight lag, peak arrival shift after alignment
Frequency	FFT-band energy, centroid, spectral change
Time-frequency	Wavelet-energy descriptors where computationally justified

A preliminary path-specific residual-energy index can be written as DI = ||x_test - x_ref||_2 / (||x_ref||_2 + epsilon). The operation presumes compatible scaling and time alignment. It is a change indicator, not a direct estimate of crack size or severity.

06 Proposed novelty algorithm: TASP

Temperature-Aware Sensor-Path Prioritization (TASP)

We propose a transparent, train-derived path-and-frequency ranking score. For each candidate path p and frequency f, D(p,f) quantifies reference-damage discriminability using training-only labels. V(p,f) quantifies variability of the corresponding healthy features across training temperatures. An exploratory score is:

Q(p,f) = D(p,f) / [V(p,f) + epsilon]. Final ranking will also consider path redundancy, path reliability and possibly the number of unique transducers required.

Possible definitions include a robust median-separation effect size for D and a temperature-conditioned median absolute deviation for V. Because the ratio can reward unstable low-variance estimates, all components will be normalized, uncertainty-checked, and ablated; simple alternatives (e.g. additive weighted ranking) will be tested.

Greedy constrained selection

16. Compute candidate features and path rankings using training acquisitions only.
17. Select the highest-scoring candidate and penalize redundant correlation with already selected paths.
18. Repeat until the chosen path budget K or unique-transducer budget is reached.
19. Use a nested validation scheme to tune K, weighting coefficients, and model hyperparameters.
20. Freeze the selected path list before evaluating each held-out fold; rebuild from scratch for the next fold.

Mandatory comparisons (to establish incremental value)

Baseline	Purpose
All paths, fixed features	Upper-resource reference; may not be best under shift
Random K paths (many seeds)	Checks improvement beyond path-number reduction
Damage-only discriminability	Tests contribution of thermal stability term
Thermal-stability-only ranking	Tests whether stability alone is sufficient
Conventional feature selection	Compare to e.g. mutual information / RFE, if time permits
TASP without redundancy penalty	Ablates non-redundancy term

The TASP name is an internal project label, not a claim that the mathematical idea is unprecedented. Publication-level novelty requires a deeper prior-art search and experimental superiority.

07 Missing-measurement and sensor-failure robustness

Two levels of simulated failure will be evaluated on recorded data. (A) Path dropout removes random or strategically selected path/frequency records. (B) Transducer outage removes every path where a selected transducer acts as transmitter or receiver. This distinction is essential: fewer paths do not necessarily mean fewer physical sensors.

Scenarios

Scenario	Method	Question answered
Random path loss	Mask 10%, 25%, 50% of selected paths	Graceful degradation under sporadic missing records?
Single-sensor outage	Remove all paths involving one transducer	Tolerance to one unavailable physical transducer?
Worst-case path loss	Remove top-ranked paths first	Dependence on narrow critical paths?
Train-time path masking	Apply masks only inside training partitions	Does mask-aware training improve fault tolerance?

Missing paths will be represented explicitly as unavailable rather than silently replaced with fabricated healthy measurements. Strategies may include mask features, aggregate statistics over available paths, and a fallback selector that uses pre-ranked substitutes. All substitutes are chosen without access to test labels.

08 Reliability-aware decision logic

Baseline model family

The primary classifier will use interpretable features with Logistic Regression and Random Forest or SVM benchmarks. One-class healthy-reference anomaly detection may be included as a robustness baseline, particularly for unseen damage positions. Large neural networks are not required for the core project.

Calibration and selective prediction

Classification probabilities will be checked using Brier score, reliability diagrams, and calibration error. If supported by adequate validation data, calibrated probabilities will be obtained using a reserved calibration fold. Thresholds for issuing an automatic decision will be fixed from calibration data only; uncertain outputs become Inconclusive, without using ground truth from the inspected case.

Software output	Decision interpretation
No reference damage detected	Selected test signal is classified as healthy within validated domain.
Reference damage indicated	Signal resembles experimental reversible reference-damage conditions.
Inconclusive / review required	Uncertainty, missing paths or out-of-domain conditions prevent a reliable decision.

Proposed guardrails

- Display temperature, frequency, available path count, and test-domain coverage alongside the result.
- Warn when the temperature is outside the validated range or when an unknown frequency/sensor ID is provided.
- Do not equate an SVM probability or model confidence with physical certainty of damage.
- Use selective risk-versus-coverage evaluation: a lower error rate is meaningful only with its accompanying abstention rate.

09 Evaluation protocol and reproducible experiments

Splitting policy and the confounding problem

All paths and frequencies originating from one underlying acquisition must belong to the same partition. Consecutive temperatures within a recording cycle are correlated, so simply shuffling individual waveforms is prohibited. Temperature-band holdout (e.g. contiguous ranges) and cycle-conscious analysis are preferred. With only one damaged cycle per reference location, a strict across-cycle test may be inseparable from a change in damage position; the report will explicitly separate what is identifiable from what is confounded.

Leave-one-damage-location-out analysis is planned only as a stress test: supervised ranking and classifier fitting must exclude that position entirely. Success in this setting cannot prove localization performance on actual delaminations or independent specimens.

Experiment matrix

ID	Experiment	Primary output
E0	Manifest, provenance and class-balance audit	Data audit table; reproducible partitions
E1	All-path signal index and classical ML baselines	F1, recall, false-alarm rate, ROC-AUC where valid
E2	Blocked temperature generalization	Performance vs. held-out temperature band
E3	TASP vs random / damage-only / stability-only	Performance vs. number of paths and unique sensors
E4	Dropout and simulated single-sensor outage	Robustness curves and failure-case analysis
E5	Calibration and abstention threshold	Brier, calibration diagram, risk–coverage
E6	Leave-one-reference-damage-position-out	Exploratory transfer result; cautious interpretation

Reporting metrics

Classification: balanced accuracy, macro F1, reference-damage recall, healthy false-positive rate, confusion matrices and ROC-AUC/PR-AUC where applicable. Calibration: Brier score, calibration curve and expected calibration error with justified binning. Sensor/path efficiency: number of selected paths, unique transducers, feature extraction time, model inference time. Reliability: selective risk, coverage and robustness degradation. Uncertainty: bootstrap confidence intervals resampled at the appropriate acquisition/cycle group level; dependence limitations will be disclosed.

What success looks like

Mandatory project success means reproducing the experimental protocol, completing baseline comparisons, showing transparent plots and delivering a functional application even if TASP does not improve accuracy. A scientific improvement claim requires measured benefit on held-out groups at equal path/sensor budget with uncertainty and counterexamples reported. No numeric target or expected accuracy is fabricated at proposal stage.

10 System architecture and demonstration



Figure 2. Proposed software pipeline. All methods, comparisons and outcomes shown are planned; this is not a results figure.

Proposed software stack

Layer	Selection
Data loading	Python, h5py, NumPy, pandas
Signal processing	SciPy (filters, FFT, cross-correlation), optional PyWavelets
Models and calibration	scikit-learn
Ranking and optimization	NumPy / SciPy + reproducible custom TASP module
Visualization	Matplotlib / Plotly
User interface	Streamlit running locally or via an available free-hosting option
Version control and testing	Git, pytest, configuration files, fixed random seeds

Dashboard modules

- Dataset explorer: browse scenario, temperature, frequency, transmitter and receiver.
- Signal diagnostics: healthy/reference waveform overlay, FFT, windowed residuals, damage-index plots.
- Sensor efficiency: selected-path map, path-frequency score table and budget-performance plots.
- Fault simulator: disable particular paths or an entire transducer, rerun diagnosis and track coverage.
- Reliability tab: calibrated probability, confidence warning, reason for abstention and diagnostic report export.
- Research tab: compare models using locked test partitions and display reproducibility metadata.

Sample faculty demonstration

21. Choose a held-out public acquisition and visualize its received guided-wave traces.
22. Compare ordinary all-path inference with a TASP-ranked reduced-path configuration.
23. Change the virtual availability of transducers and observe failure or abstention behavior.
24. Display the decision, calibration status and validation-domain notes.
25. Export an inspection-style PDF/CSV report with model and dataset provenance.

11 Work plan, milestones, and dependencies

Week	Scope	Acceptance checkpoint
1	Acquire/audit OGW #2; build HDF5 reader, metadata manifest and splits.	Integrity audit plus first valid waveform plots.
2	DSP pipeline and baseline damage indices/classifiers.	Reproducible baseline metrics without leakage.
3	Implement TASP ranking and path-budget studies.	Ablations and budget plots on reserved validation.
4	Develop path dropout, transducer outage and mask-aware strategies.	Failure stress-test matrix and curves.
5	Calibrate scores; selective prediction and grouped robustness tests.	Calibration/risk–coverage and held-out evaluation.
6	Build dashboard, interpret findings, package code and final report.	Live reproducible demonstration and dossier-to-results traceability.

Baseline and stretch scope

Committed minimum	Conditional advanced extension
Temperature-band evaluation, TASP ranking, path budgets, single-transducer dropout, selective prediction, Streamlit dashboard.	Leave-one-location-out tests, anomaly-detection benchmark, additional OGW dataset transfer, expanded uncertainty analyses.

Dependencies

Existing laptop and stable internet for dataset download; Python 3.11+ environment; adequate local disk for a selected OGW subset (full archive size to be confirmed); no sensors, test chamber, composite sample, Arduino, DAQ, finite-element software license or paid cloud GPU required. A GPU is optional, not part of core scope.

12 Budget and feasibility

Category	Expected outlay	Notes
Original experimental data	INR 0	Publicly accessible OGW dataset; download/connection costs not estimated.
Python and numerical libraries	INR 0	Open-source stack; verify package licenses before redistribution.
Sensors, specimen, laboratory equipment	INR 0	Existing experimental records only.
Computing	INR 0 incremental	Assumes access to an existing capable laptop.
Report and demonstration software	INR 0	Local Streamlit and standard office exports.

Estimated direct project materials/software cost: INR 0 under existing-laptop and internet assumptions. Disk, data transfer, electricity and optional printing are not quantified.

13 Risks, limitations and mitigations

Risk / limitation	Planned control
Large/complex HDF5 archive	Start with 1-2 frequencies; verify file schema and use chunked loading/caching.
Cycle/session confounding	Grouped acquisition splits, restricted conclusions, explicit experimental caveats.
Proxy defect is not delamination	Use “reference damage” in UI/report; no internal-delamination claims.
Insufficient independent data for calibration	Reserve calibration set, use simpler methods, show wide uncertainty; disable unsupported outputs.
Temperature-selection overfitting	Nested grouped validation; rank paths within training folds only.
Robustness under unseen conditions not established	Treat OOD alert as warning/diagnostic, not proof of robustness.
TASP fails to beat standard methods	Publish negative results, ablations and cost/error trade-offs; retain engineering contribution.
Overextended project scope	Lock baseline + core E0-E5; delay localization and extra datasets.

14 Deliverables and faculty evaluation plan

Submission package

26. A version-controlled Python project with documented environment setup and preprocessing scripts.

27. Immutable dataset manifest and checksum list; no redistribution of raw files unless permitted.
28. Implementation of TASP ranking, conventional baselines, robustness simulation and confidence calibration.
29. Reproducible experiment configuration and separated train/validation/test acquisition-group IDs.
30. Interactive WaveSense-CFRP dashboard with inspect/compare/failure/calibration views.
31. Final report with tables, figures, all negative or positive results, limitations, acknowledgments and citations.
32. Faculty presentation with live demo and one-page experimental summary.

Faculty acceptance checklist

☐ Research question and constraints approved

☐ OGW Dataset #2 access/schema verified

☐ No ground-truth leakage in any model-selection step

☐ TASP compared against random and conventional ranking

☐ Temperature band and transducer outage evaluated

☐ Calibrated / inconclusive output documented

☐ Results separated from proposed capabilities

☐ Original data creators and prior art cited

Project approval

Faculty guide	Student / team representative
Name: _____	Name: _____
Signature: _____	Signature: _____
Date: _____	Date: _____

Appendix A Implementation structure and reproducibility

Recommended repository layout (adaptable after HDF5 schema inspection):

- data_manifest/ - provenance, checksums, split IDs and dataset notes
- src/io/ - HDF5 ingestion and group metadata validation
- src/dsp/ - filtering, baseline construction and engineered features
- src/tasp/ - train-only temperature-aware path ranking
- src/models/ - baselines, estimators, confidence calibration
- src/stress/ - path dropout, sensor-outage and blocked evaluation
- app/ - Streamlit visualization, classification and report view
- experiments/ - fixed YAML configurations, scripts and saved metrics
- tests/ - metadata, no-leakage and dropout behavior tests

Data-leakage prevention checklist

- Group raw channels from the same acquisition before creating partitions.
- Fit scalers, references, selected features and TASP scores using training data only.
- Tune hyperparameters and abstention thresholds on validation/calibration data only.
- Reserve final held-out groups until the pipeline and selection rules are frozen.
- Log random seeds, code commits, software versions and exact dataset identifiers.
- Use consistent counting unit (acquisitions/cycles vs waveforms) when reporting sample sizes.

Appendix B Evidence inventory and dataset audit template

Evidence claim	Source / required check	Status
12 transducers; frequency and temperature specification	Scientific Data 2019, Methods	Documented
Four reversible surface-disc positions	Scientific Data 2019, Methods	Documented
Two healthy cycles; one damaged cycle per location	Scientific Data 2019, Data Records	Documented
HDF5 fields and sampling-rate specifics	Inspect downloaded native HDF5 files	Not yet audited
Per-class independent sample counts	Acquisition manifest and grouping	Not yet audited
Actual ML performance and robustness gain	Run locked experiments E1–E6	Not measured
Generalization beyond single laboratory specimen	Independent-specimen datasets needed	Not established

References and source record

[S1] SRMIST. Fundamentals of Composite Materials (21MEO101T), course syllabus, 2021 regulations, revised July 2024. User-supplied course PDF.

[R1] Moll, J., Kexel, C., Pötzsch, S., Rennoch, M., & Herrmann, A. S. (2019). Temperature affected guided wave propagation in a composite plate complementing the Open Guided Waves Platform. Scientific Data, 6, 191.
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[R3] Croxford, A. J., Moll, J., Wilcox, P. D., & Michaels, J. E. (2010). Efficient temperature compensation strategies for guided wave structural health monitoring. Ultrasonics, 50, 517-528.
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